

House Price Analysis



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Report Overview

Executive Summary

This report presents an analysis of housing price prediction for Ames, Iowa, using machine learning techniques. It aims to provide actionable insights into the factors influencing housing prices and demonstrate the effectiveness of advanced predictive modeling.

Objective

To predict house prices in Ames, Iowa, using machine learning models and identify the key factors affecting these prices.

Methods

Implemented six advanced machine learning approaches: Ridge Regression, Lasso Regression, Elastic Net Regression, Support Vector Machines (SVM), XGBoost, Principal Component Analysis (PCA) with K-Means clustering, and Artificial Neural Networks (ANN)

Results

SVM (linear kernel) and ANN achieved the best performance with a 5-fold validation R^2 of 0.88. ANN however had a higher 5-fold RMSE at \$27,139.34 vs SVM's 5-fold RMSE of \$26,559.73. XGBoost overfitted despite extensive hyperparameter tuning. Random Forest also struggled as well.

Elastic and Ridge Regression had competitive 5-Fold R^2 of .86 along with insights into what factors affected the predictions.

Insights

Models like Elastic Net Regression are interpretable and suitable for actionable insights, while SVMs and Neural Networks offer robust prediction capabilities.

Data Cleaning

Dataset Overview

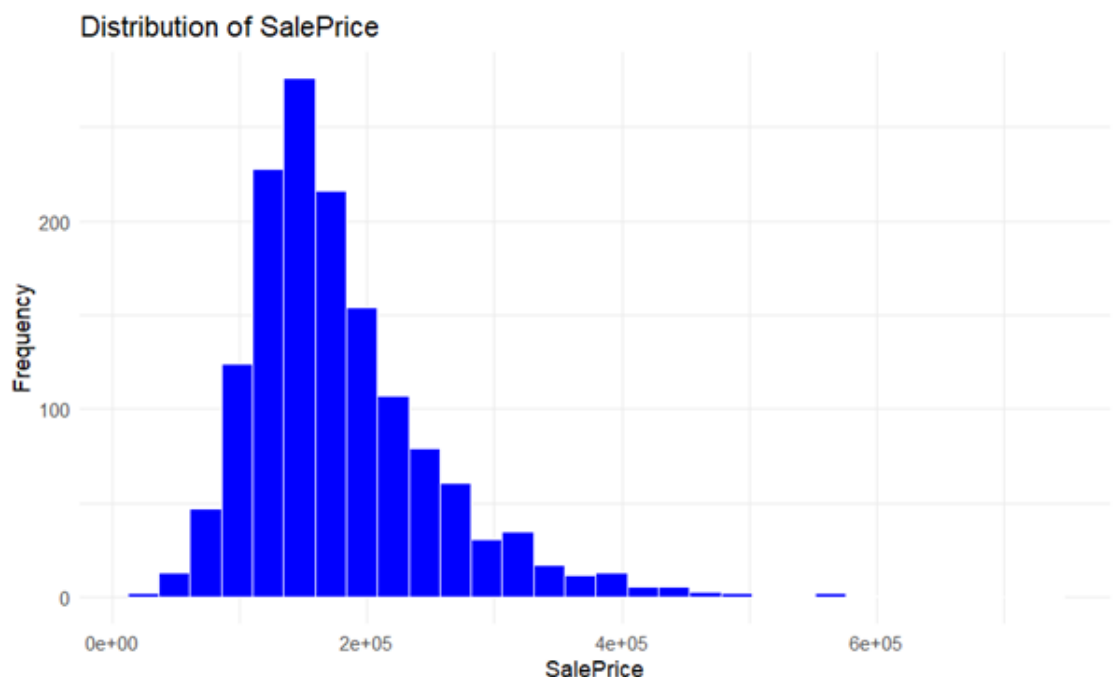
The dataset is derived from the House Prices - Advanced Regression Techniques competition. Initial data cleaning reduced observations from 1,460 to 1,383 after handling missing values and outliers.

Data Preprocessing Steps

Handle missing values using mean imputation and domain-specific rules. For instance, there is a variable called “Fence” which contains missing values. The missing values however just mean that there is “No Fence”. Thus, the variable was recoded to “No Fence”. There are several variables like this where “NA” indicates that the features does not exist for that particular house.

Afterwards, there were some variables like “Lot Frontage” which were numeric and had missing values. These were imputed with the median while a few categorical variables were imputed with the mode. Finally, the Local Outliers Algorithm was utilized to deal with any outliers.

The graph on the right shows the target variable's distribution: SalePrice before cleaning



Ridge and Elastic Regression

Models Overview

Elastic Net Regression combines Lasso and Ridge (L2) penalties, making it effective for high-dimensional data and correlated predictors. Lasso Regression is useful for multi-collinearity. Ridge Regression is a linear regression technique that addresses multicollinearity by adding an L2 penalty to the loss function. This penalty shrinks coefficients toward zero but does not eliminate them, reducing overfitting and improving model generalization. It is particularly effective when predictors are highly correlated.

Model Performance

Elastic Regression:

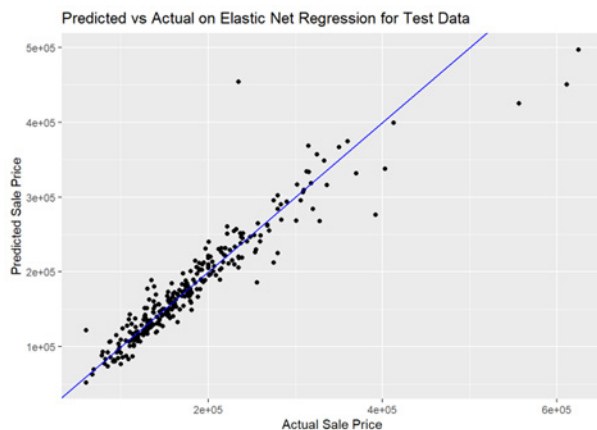
5-fold R^2 : 0.86 – The model explains 86% of the variance in house prices.

5-fold RMSE: 29,316.87 – The predictions for house prices are off by an average of 29,316.87 dollars.

Ridge Regression:

5-fold R^2 : 0.86 – The model explains 86% of the variance in house prices.

5-fold RMSE: 28,639.64 – The predictions for house prices are off by an average of 29,316.87 dollars.



This model is doing a fairly good job with most of the data points closely clustered around the blue line. There are only a few data-points that deviate a lot.

The top 5 predictors from both models are:

NeighborhoodStoneBr: 25,864.54

NeighborhoodNoRidge: 21,469.17

NeighborhoodNridgHt: 21,220.20

RoofMatlMembran: 32,458.18

Condition2PosN: -124,602.6

SVM & ANN

Model Performance

Support Vector Machine (SVM):

5-fold R^2 : 0.88 – The model explains 88% of the variance in house prices.

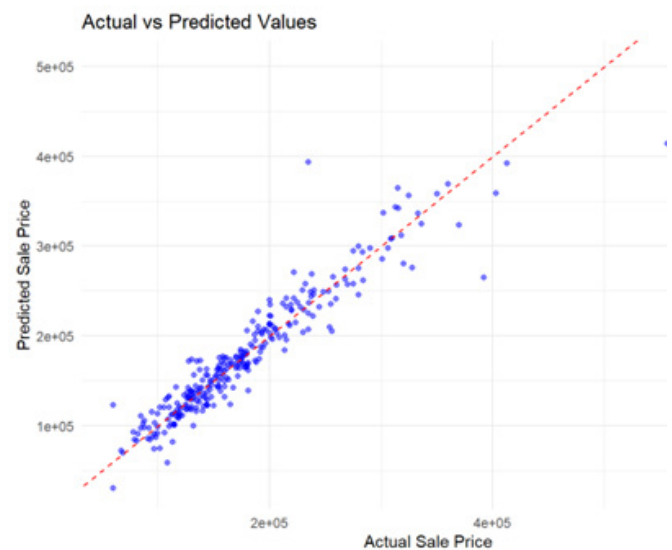
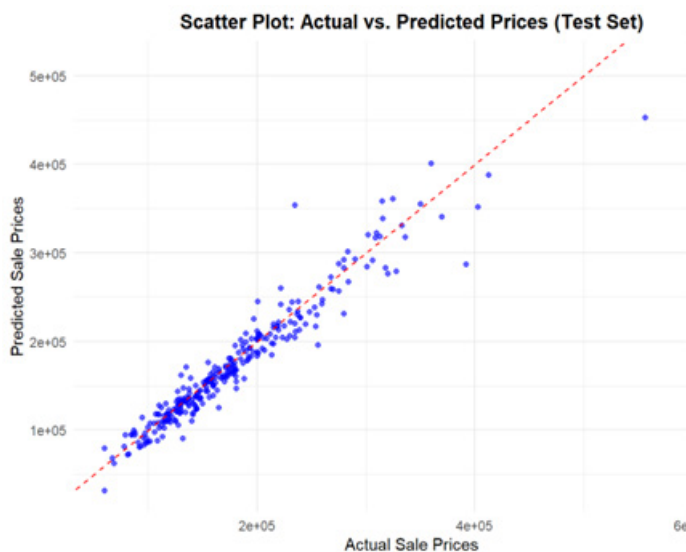
5-fold RMSE: \$26,559.73 – The predictions for house prices are off by an average of 27,139.34 dollars.

Artificial Neural Network (ANN):

5-fold R^2 : 0.88 – The model explains 88% of the variance in house prices.

5-fold RMSE: \$27,139.34 – The predictions for house prices are off by an average of 26559.73 dollars.

Both of these models have similar performance but SVM is better with a lower RMSE. They are additionally black box models which makes getting insights difficult. These models however do the best at prediction.



The scatter plot on the left is for ANN and the Scatter plot on the Right is for SVM . Both scatter plots show that the models are doing a really great job at predicting the values. There are very few points that deviate away. Thus, SVM should be kept for its predictive power.

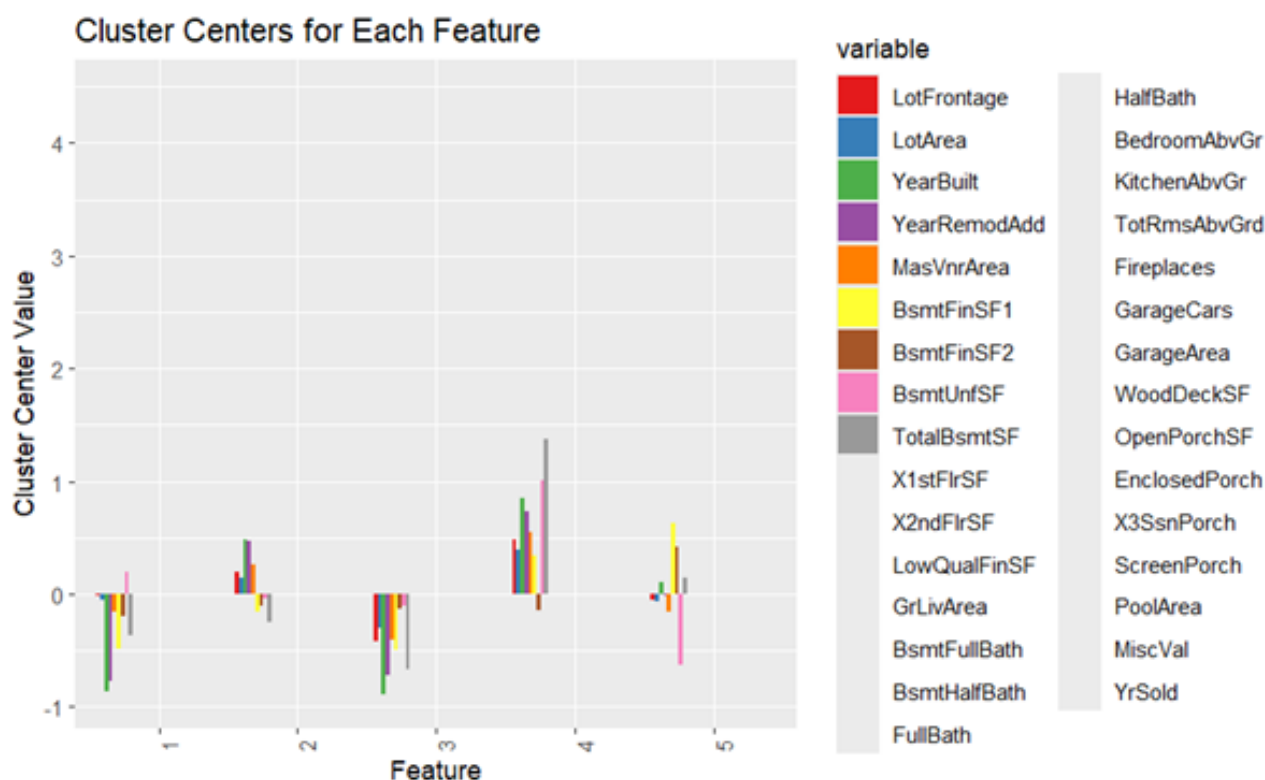
PCA & Clustering

Overview

PCA was first performed to reduce the dimensionality of the dataset. Afterwards clustering with the elbow method was performed.

Insights

PCA was first performed to reduce the dimensionality of the dataset. Afterwards 5 k-mean clusters were created.



This graph helps to visualize the most important features of the clusters. A larger bar in either direction indicates that the feature plays an very influential role in differentiating the clusters. The bars that are colored represent the largest difference. In the context of Ames, Iowa, the basement variables followed and “Lot” variables have a big influence. An interesting thing is that all these variables describe area in some form or other which suggests that “area” has an influence on the clusters. After that “YearRemodAd” and “Year Built” have an influence which suggests that the age of the home and rennovating the home play another influence in the clusters.

Conclusions

Summary

Neighborhood, Area related variables, etc. are important predictors of house prices. Thus, when pricing houses, these are the key variables that we should focus on.

Limitations

This data is only limited to the Ames, Iowa neighborhood which means that results cannot be necessarily generalized to the entire United States.

Additionally, this is observational data which means that everything is an association. Causation cannot be established.

These are the 2 biggest limitations of the analysis

Left Out Models

XGboost and Random Forest were left out due to overfitting and being very computationally intensive to train.